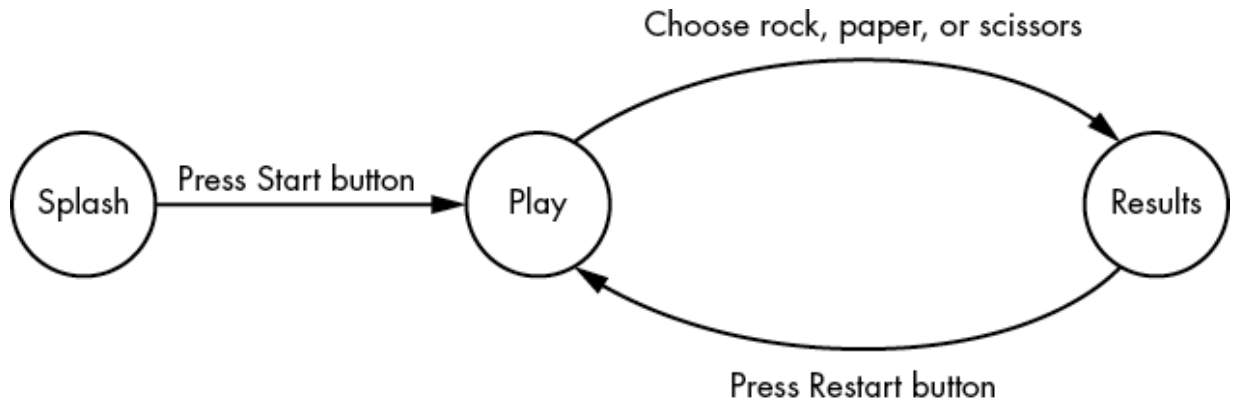




*Figure 15-4: Rock, Paper, Scissors state diagram*

When sitting idle (waiting for the user), the current scene will typically remain unchanged. That is, inside the main event loop, the program will usually not change the value of the `state` variable. (The state might change when a timer ends, but this will be rare.) This game starts in the Splash scene, and when the user presses the Start button, the game moves to the Play scene. The play of the game then alternates between the Play and Results scenes. Though this is a simple example, a state diagram can be very useful in understanding the flow of more complicated programs.

[Listing 15-1](#) provides the code of the Rock, Paper, Scissors program, with the boilerplate code omitted to save space.

### **File: RockPaperScissorsStateMachine/RockPaperScissors.py**

---

```
# Rock, Paper, Scissors in pygame
# Demonstration of a state machine

--- snip ---

ROCK = 'Rock'
PAPER = 'Paper'
SCISSORS = 'Scissors'

# Set constants for each of the three states
STATE_SPLASH = 'Splash' ❶
STATE_PLAYER_CHOICE = 'PlayerChoice'
STATE_SHOW_RESULTS = 'ShowResults'

# 3 - Initialize the world
```